

Moments



Starter

1. What unit do we measure forces in?

Newtons, N

2. What unit do we measure mass in?

Grams, g or Kilograms, Kg

3. What unit do we measure weight in?

Newtons, N

4. What happens if forces are unbalanced?

The object will change speed, shape or direction

5:00

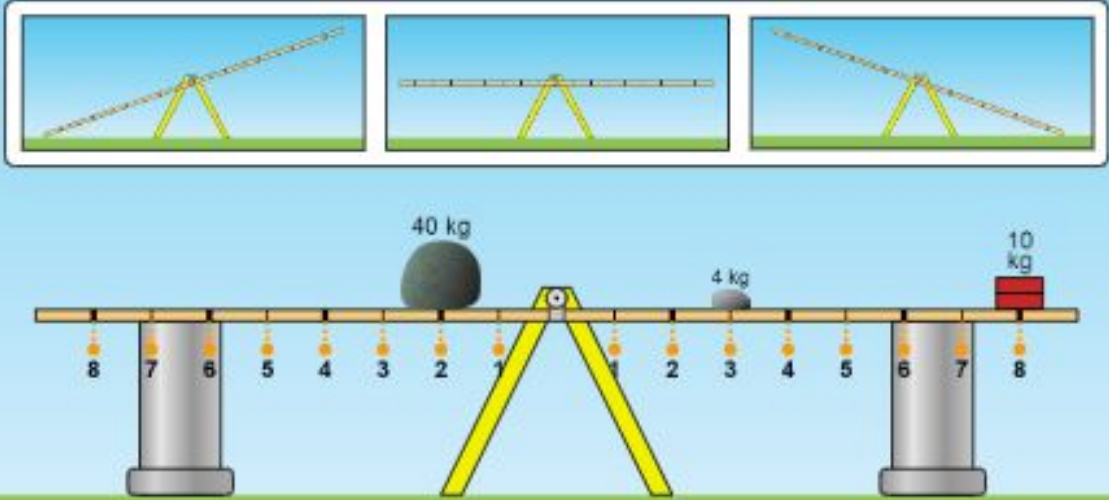
Success criteria

Lesson Title	Working towards	Working at	Working beyond
Recap: Forces & Motion (Y7)	I can identify different forces acting on an object.	I can describe what balanced and unbalanced forces are.	I can explain how unbalanced forces change an object's motion using force diagrams.
Speed	I can identify the correct units for speed, distance, and time.	I can use the speed = distance $\div$ time formula to calculate speed, distance or time.	I can solve speed questions using different units and rearranged formulas.
Describing Movement	I can read values from a distance-time graph.	I can describe motion based on the shape of the graph.	I can draw and explain distance-time graphs from written data or scenarios.
Turning Forces	I can give examples of things that turn when a force is applied.	I can use the moment formula to calculate turning forces.	I can explain how balanced and unbalanced moments affect turning.
Pressure between solids	I can state that heavier or larger objects create more pressure.	I can calculate pressure using the correct formula and units.	I can compare pressure in different situations using scientific reasoning.
Pressure in liquids and gases	I can describe how pressure changes with depth in liquids and altitude in gases.	I can explain what causes pressure in liquids and gases using particles.	I can calculate liquid pressure using the h ρ g formula and explain examples.
Particles on the move	I can define diffusion.	I can describe how temperature and concentration affect diffusion.	I can explain diffusion using the particle model and apply it to real examples.

<https://phet.colorado.edu/en/simulation/balancing-act>

What will happen?

Position
☐ None
☐ Rulers
☒ Marks



40 kg

4 kg

10 kg

8 7 6 5 4 3 2 1 1 2 3 4 5 6 7 8

Check

Level: 4 Challenge 1 of 6 Score: 0 Start Over

Balancing Act

Intro Balance Lab Game

PHET

Which screwdriver will be easier to open the can of paint with?

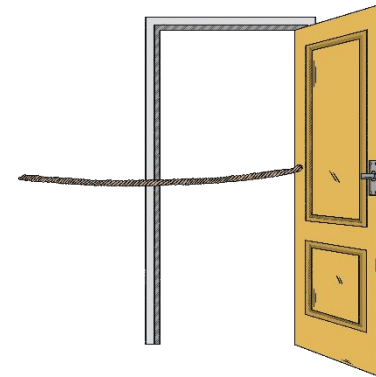


Why Does This Work?

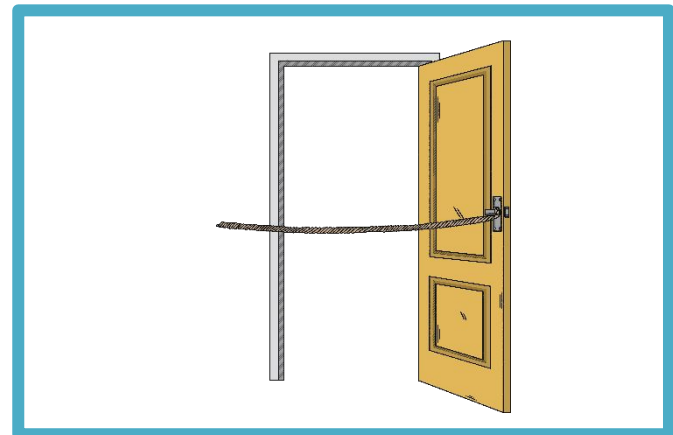
Which of the following tools makes the job easier? Can you explain why?

Job: closing an open door

tools: rope near the hinge

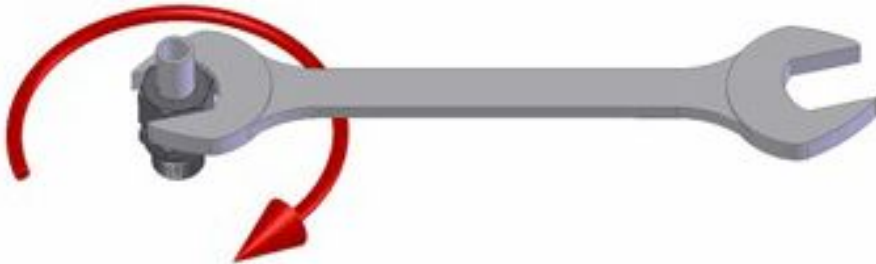
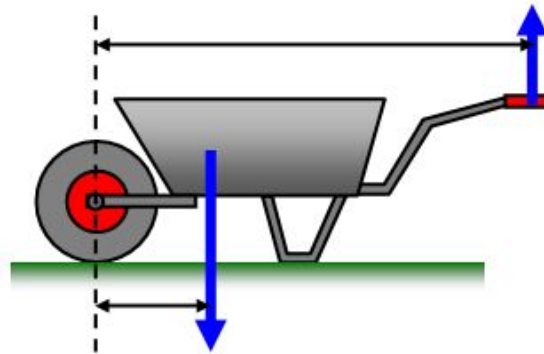


rope near the handle



Moments

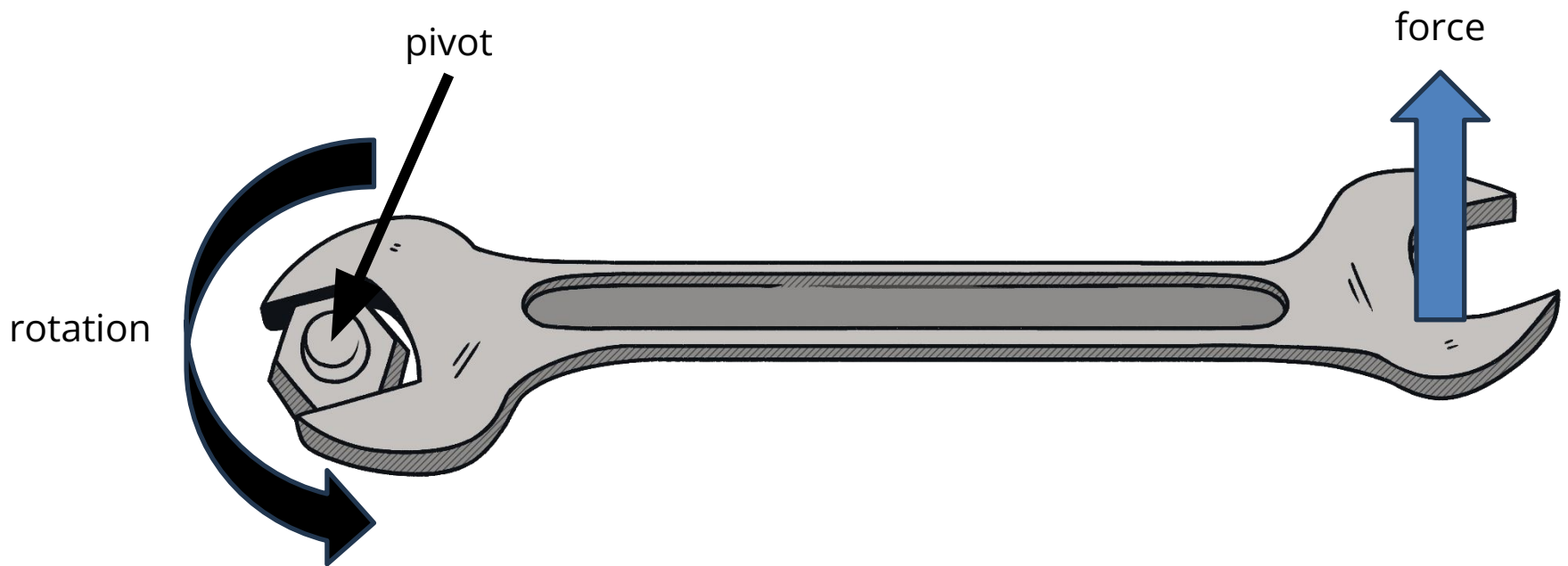
A **moment** is the turning effect of a force about a pivot.



Moments

When a force acts on an object with a **pivot**, a moment is created to cause the object to rotate about the pivot.

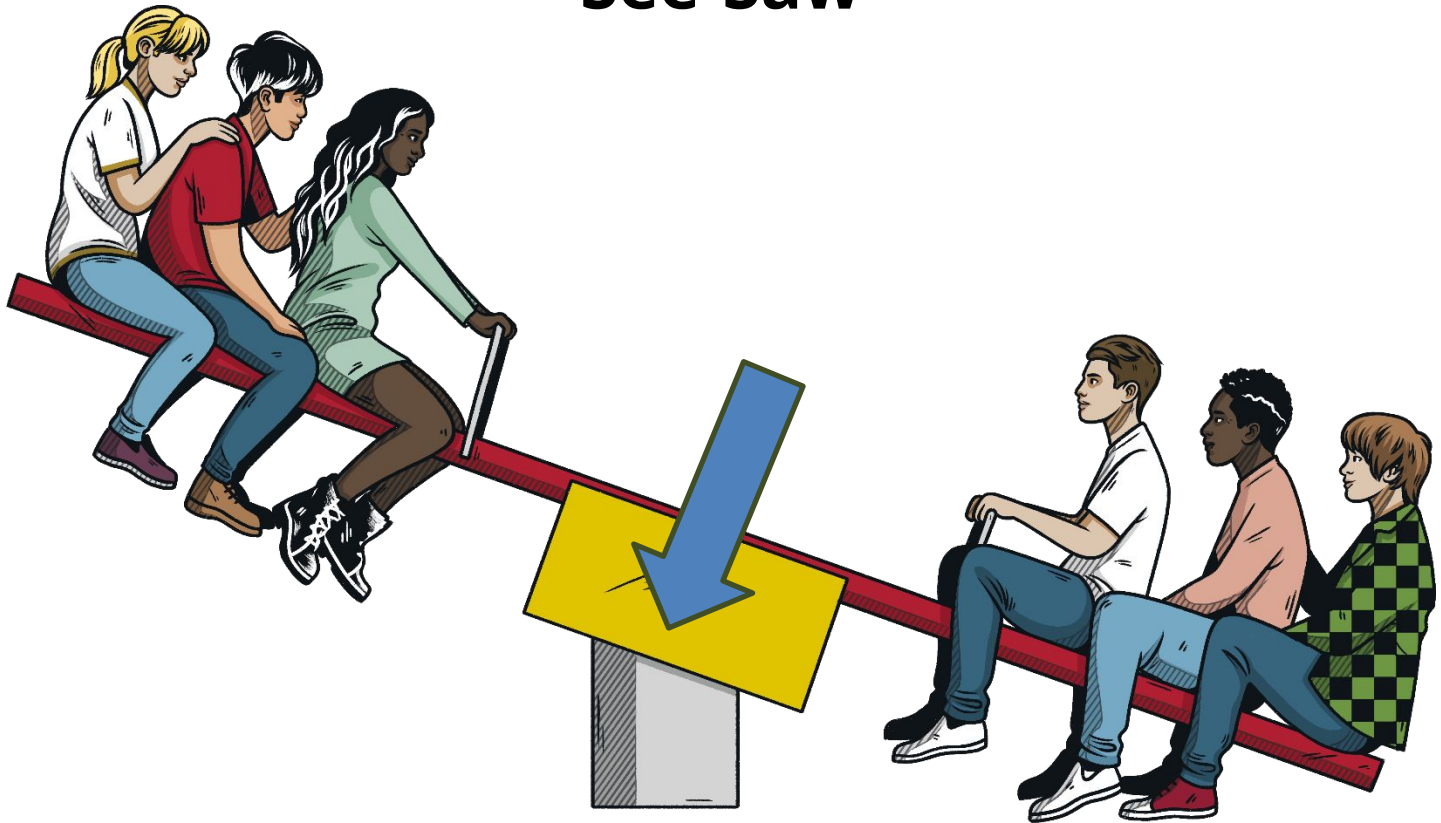
A pivot is a fixed point about which the object can rotate.



Moments

Where is the pivot on the following rotating objects?

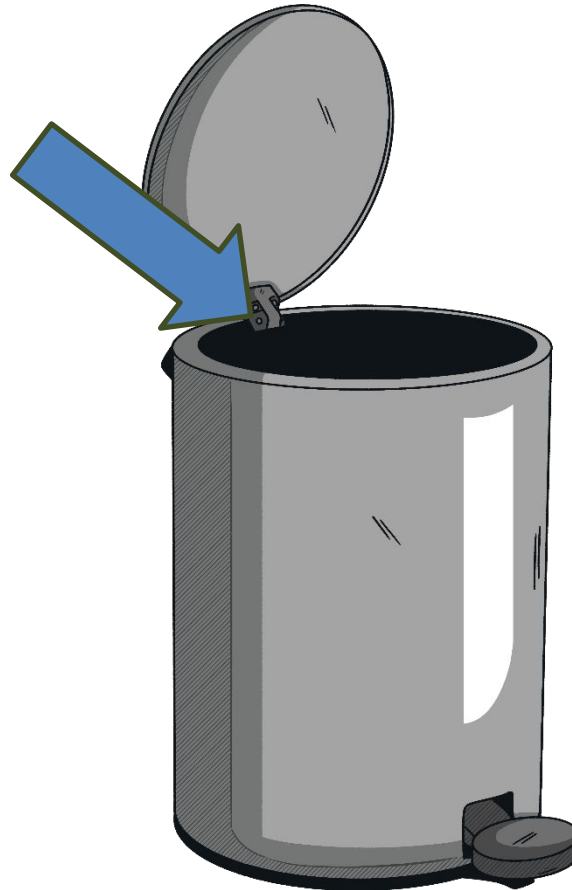
See-Saw



Moments

Where is the pivot on the following rotating objects?

lid of a pedal bin



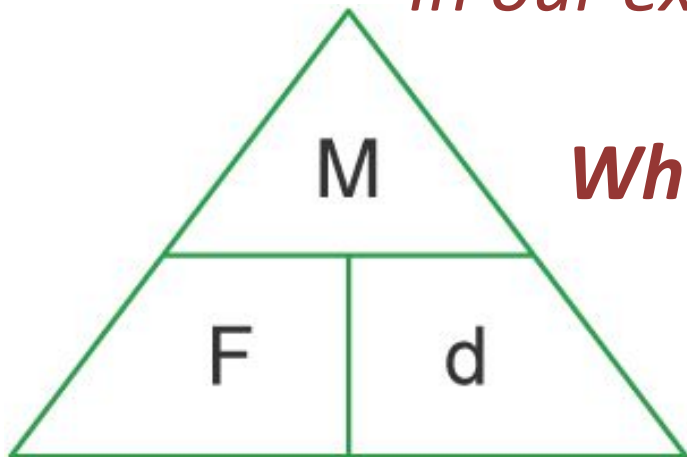
Moments



moment (....) = force applied (N) x distance (m)

$$M = Fd$$

In our experiment the force was the weight.



What must the unit of moments be?

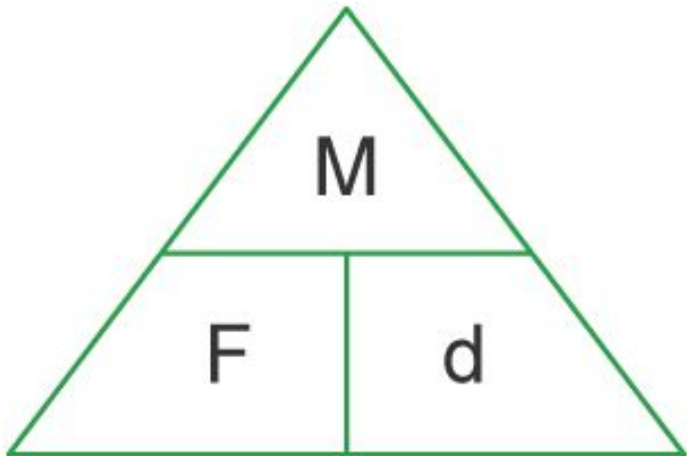
Newton metre, Nm

Moments



The size of a moment (turning force) depends on two factors:

1. The size of the force applied
2. The perpendicular distance of the force to the pivot



What do you think now / HOW CAN YOU EXPLAIN IT?

Which screwdriver will be easier to open the can of paint with?

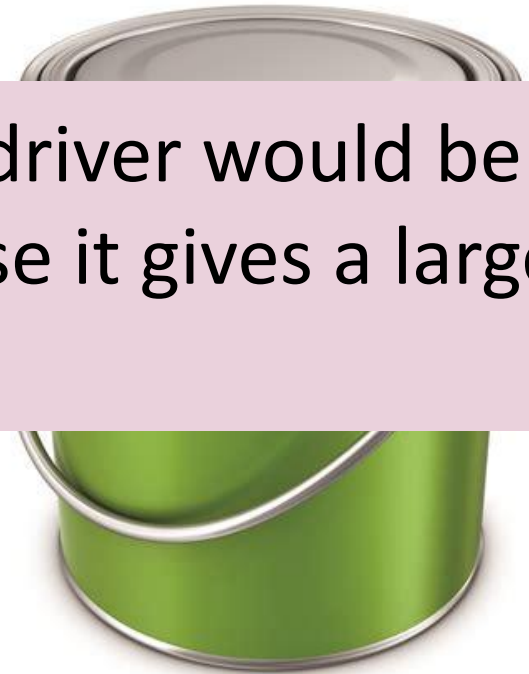


What do you think now / HOW CAN YOU EXPLAIN IT?

Which screwdriver will be easier to open the can of paint with?

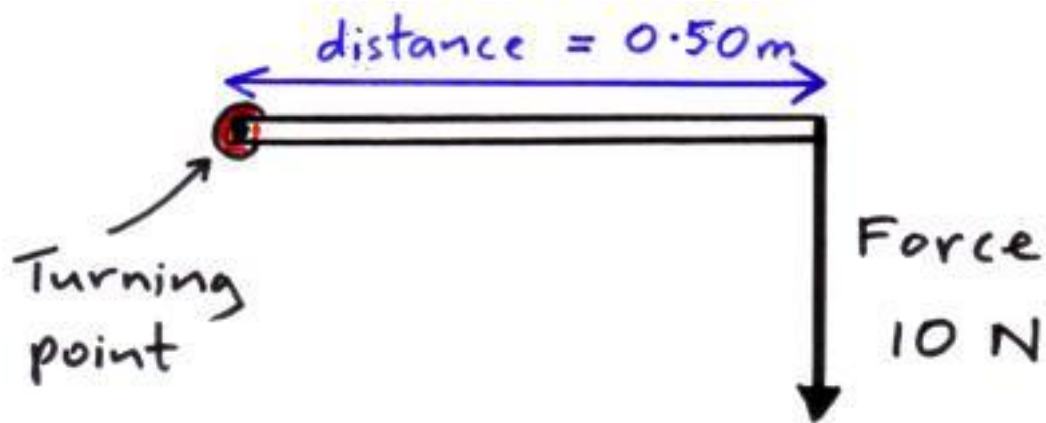


Larger screwdriver would be easier because it gives a larger moment



Example

a) Calculate the moment



$$M = F \times d$$

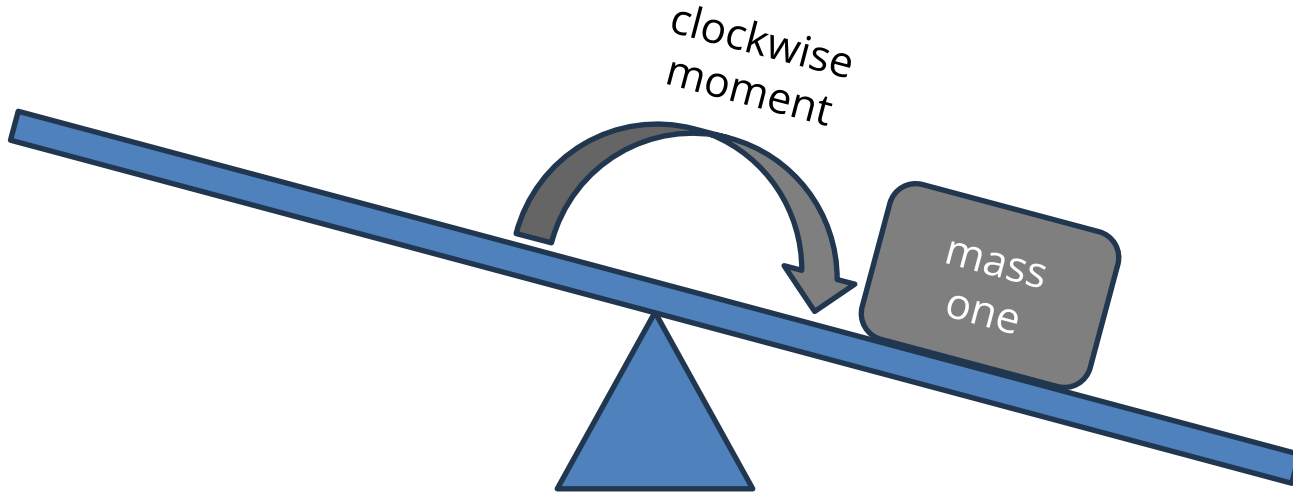
$$M = 0.5 \times 10$$

$$M = 5 \text{ Nm}$$

Principle of Moments

Consider a see-saw. If you place a mass on one side of the see-saw, it will tip.

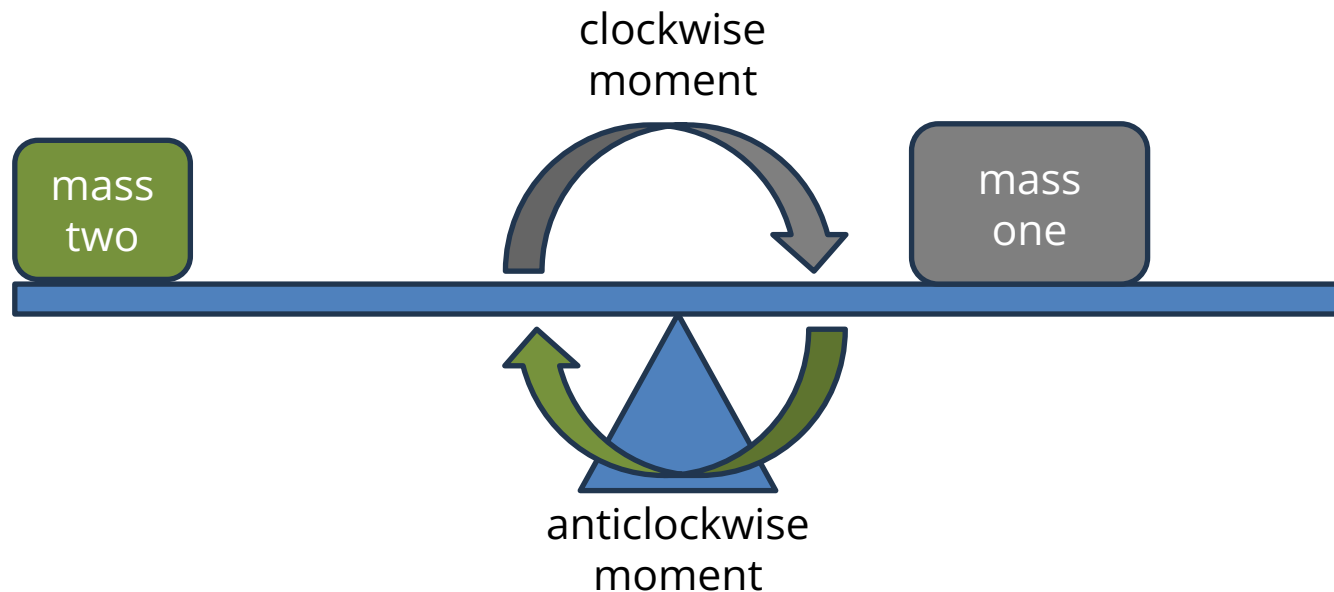
The weight of the mass causes a **resultant clockwise moment**, rotating the see-saw.



Principle of Moments

To balance the see-saw, another mass needs to be placed on the other side of the pivot.

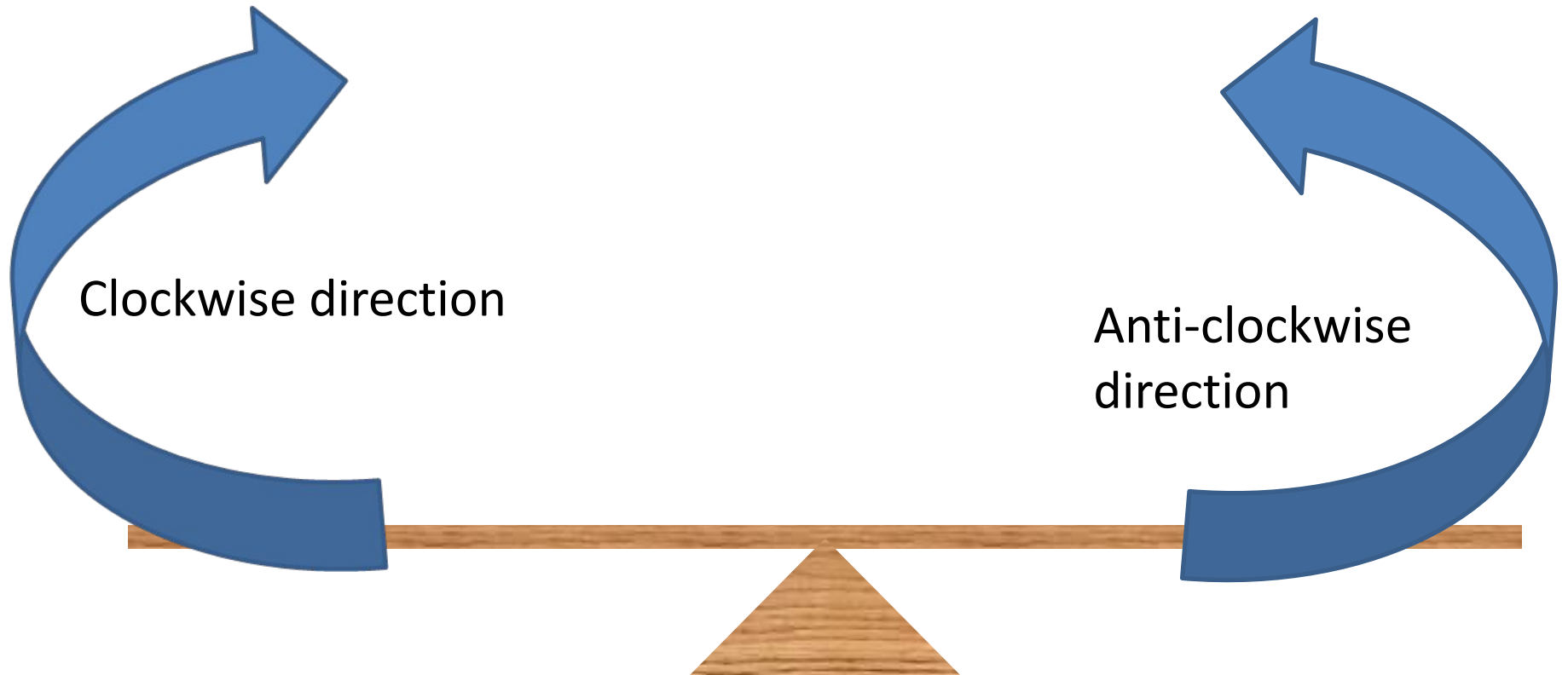
The see-saw will only balance if the **clockwise moment = anticlockwise moment**.



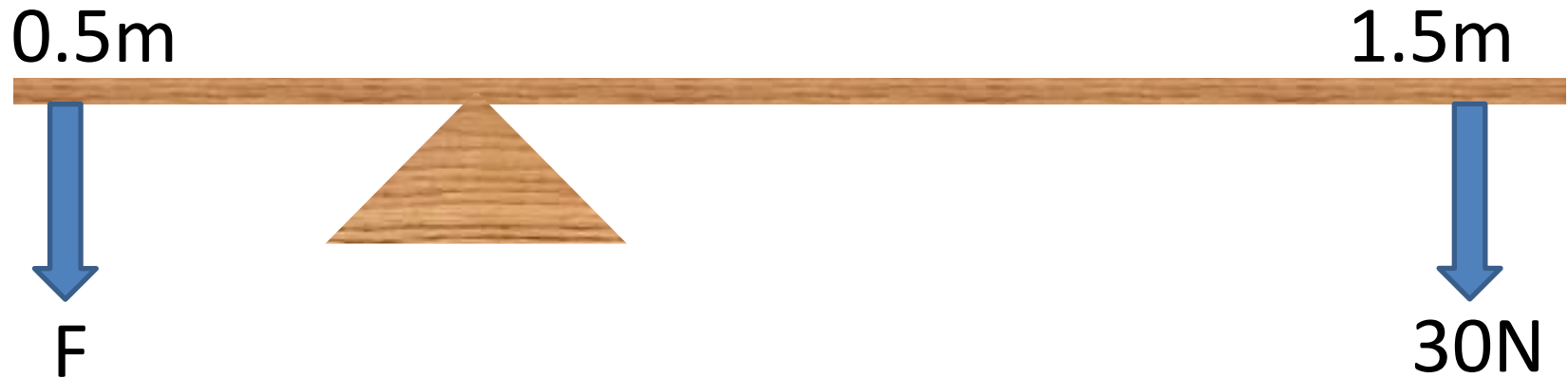
The Principle of Moments

When an object is balanced about a pivot, the total anticlockwise moment equals the total clockwise moment.

Which direction will it turn?



We can use the principle of moments to calculate an unknown distance or force



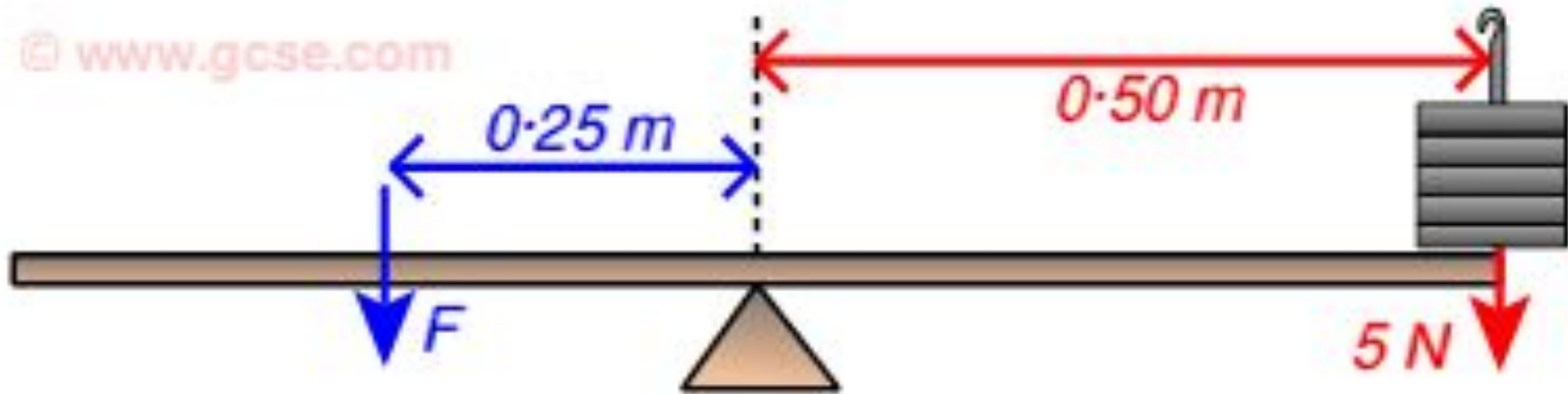
$$F \times 0.5\text{m} = 30\text{N} \times 1.5\text{m}$$

$$F \times 0.5\text{m} = 45\text{Nm}$$

$$F = 45\text{Nm} / 0.5\text{m}$$

$$= 90\text{N}$$

Calculate the Force required to keep equilibrium



$$M = F \times d$$

$$F = M/d$$

$$F = 2.5/0.25$$

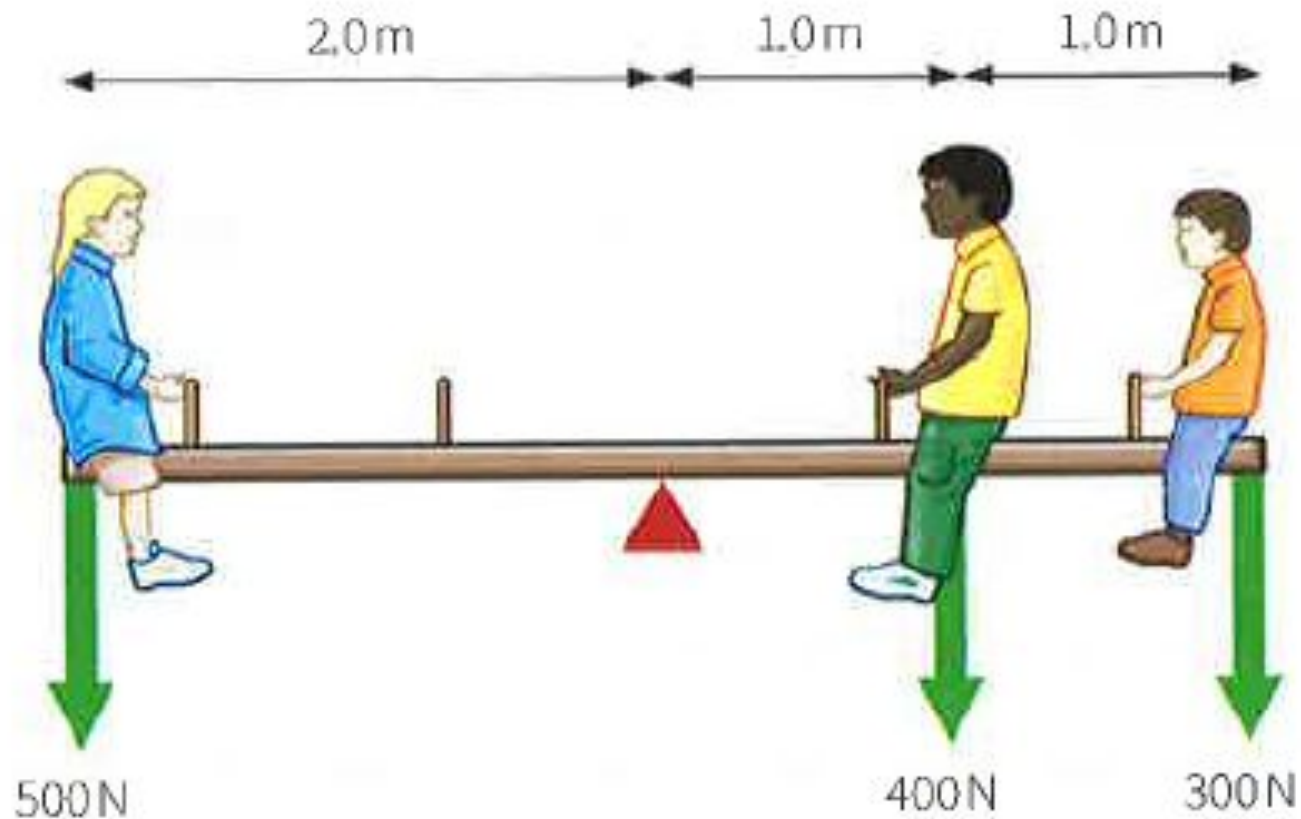
$$F = 10\text{ N}$$

$$M = F \times d$$

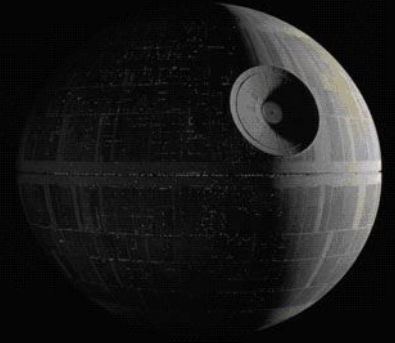
$$M = 5 \times 0.5$$

$$M = 2.5\text{ Nm}$$

Are the moments balanced?



STAR WARS SEE SAWS



Moment = Force x Distance

(Nm) (N) (m)



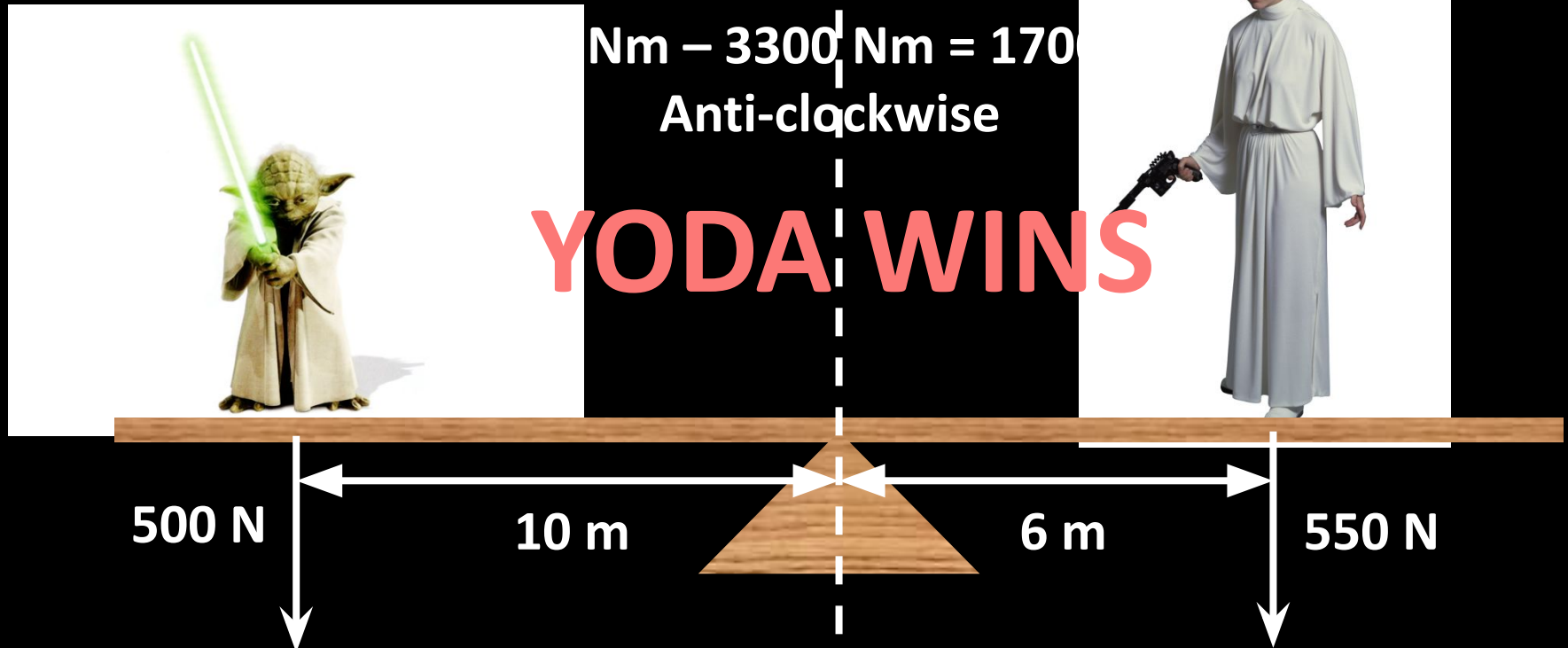
$$\begin{aligned}\text{Moment} &= 500 \text{ N} \times 10 \text{ m} \\ &= 5000 \text{ Nm}\end{aligned}$$

$$\begin{aligned}\text{Moment} &= 550 \text{ N} \times 6 \text{ m} \\ &= 3300 \text{ Nm}\end{aligned}$$

$$5000 \text{ Nm} - 3300 \text{ Nm} = 1700 \text{ Nm}$$

Anti-clockwise

YODA WINS



Round

2

Moment = Force x Distance

(Nm) (N) (m)



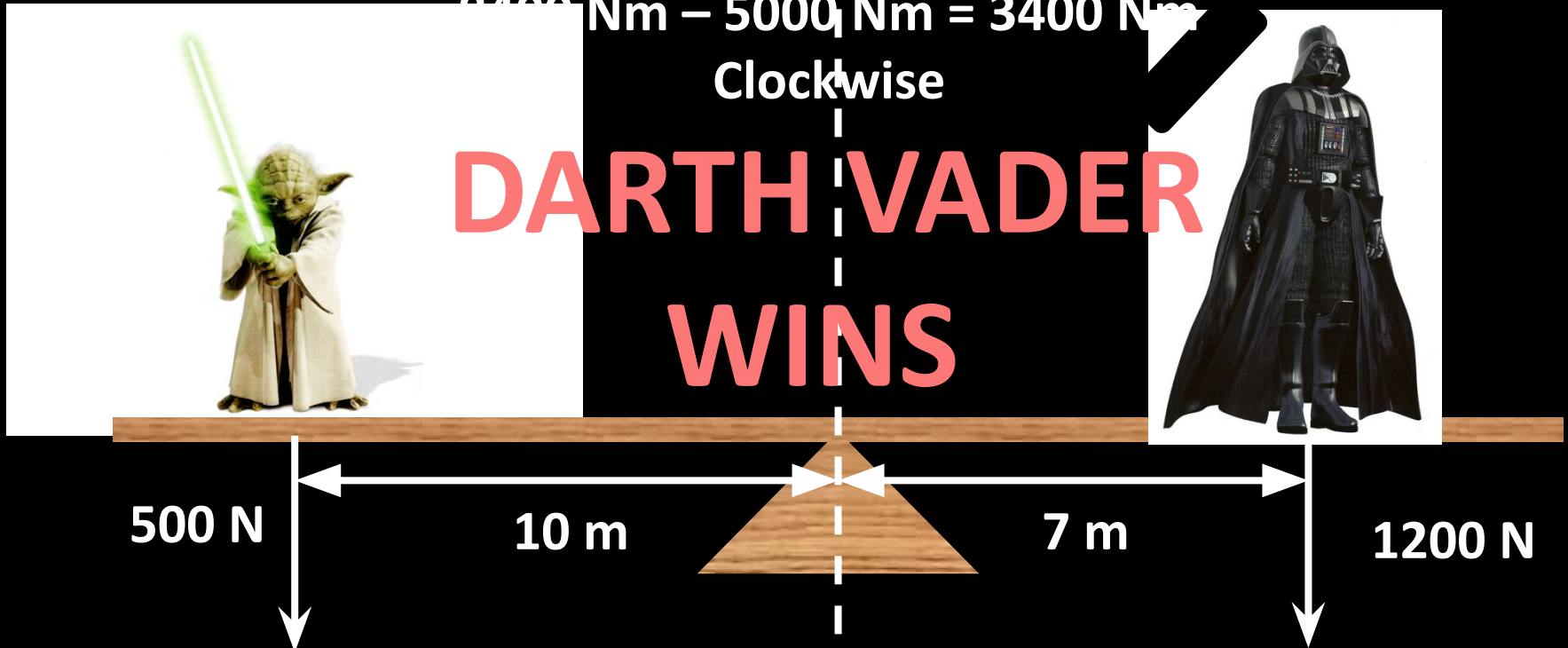
$$\begin{aligned}\text{Moment} &= 500 \text{ N} \times 10 \text{ m} \\ &= 5000 \text{ Nm}\end{aligned}$$

$$\begin{aligned}\text{Moment} &= 1200 \text{ N} \times 7 \text{ m} \\ &= 8400 \text{ Nm}\end{aligned}$$

$$8400 \text{ Nm} - 5000 \text{ Nm} = 3400 \text{ Nm}$$

Clockwise

**DARTH VADER
WINS**



Round

3

Moment = Force x Distance

(Nm) (N) (m)



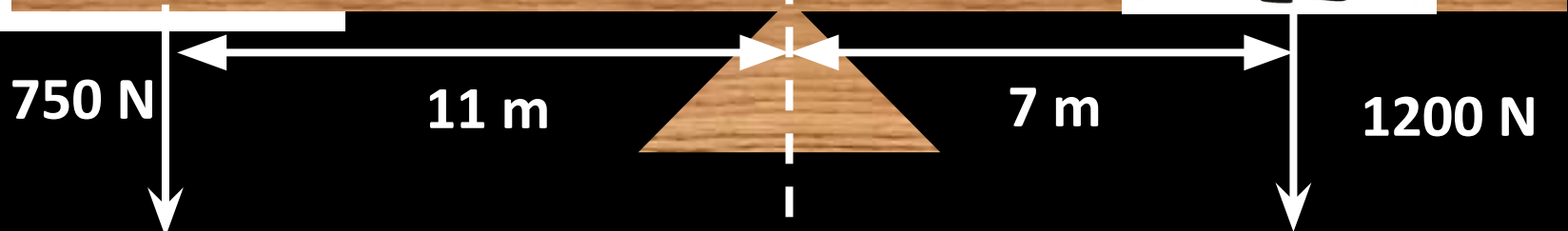
$$\text{Moment} = 750 \text{ N} \times 11 \text{ m}$$
$$= 8250 \text{ Nm}$$

$$\text{Moment} = 1200 \text{ N} \times 7 \text{ m}$$
$$= 8400 \text{ Nm}$$

$$8400 \text{ Nm} - 8250 \text{ Nm} = 150 \text{ Nm}$$

Clockwise

**DARTH VADER
WINS**



Round

4

Moment = Force x Distance

(Nm) (N) (m)



$$\text{Moment} = 3000 \text{ N} \times 8 \text{ m}$$

$$= 24000 \text{ Nm}$$

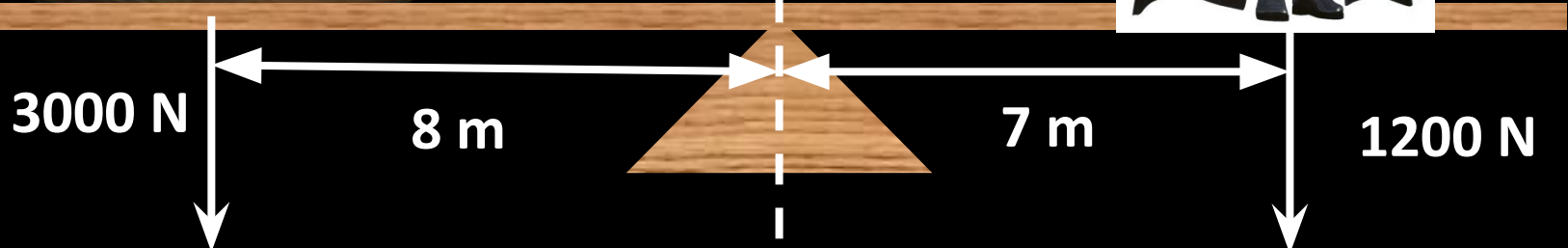
$$\text{Moment} = 1200 \text{ N} \times 7 \text{ m}$$

$$= 8400 \text{ Nm}$$

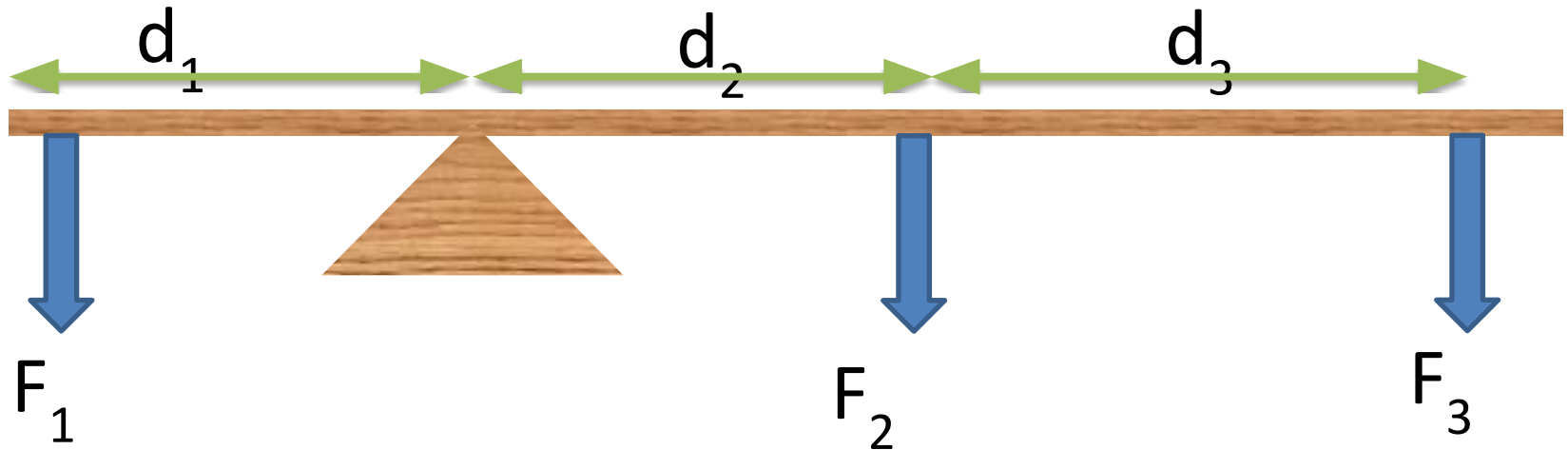
$$24000 \text{ Nm} - 8400 \text{ Nm} = 15600 \text{ Nm}$$

Anti-clockwise

JABBA THE HUTT WINS

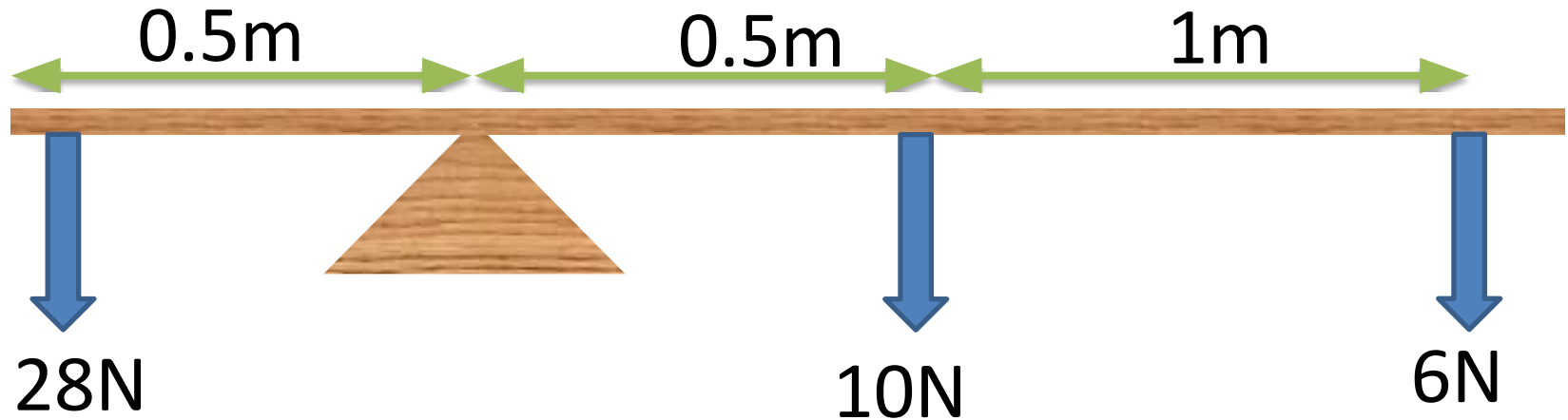


Multiple forces on a beam



- $d_1 \times F_1$
- $=M_1$
- $(d_2 \times F_2) + ((d_2 + d_3) \times F_3)$
- $=M_2$
- If at equilibrium $M_1 = M_2$

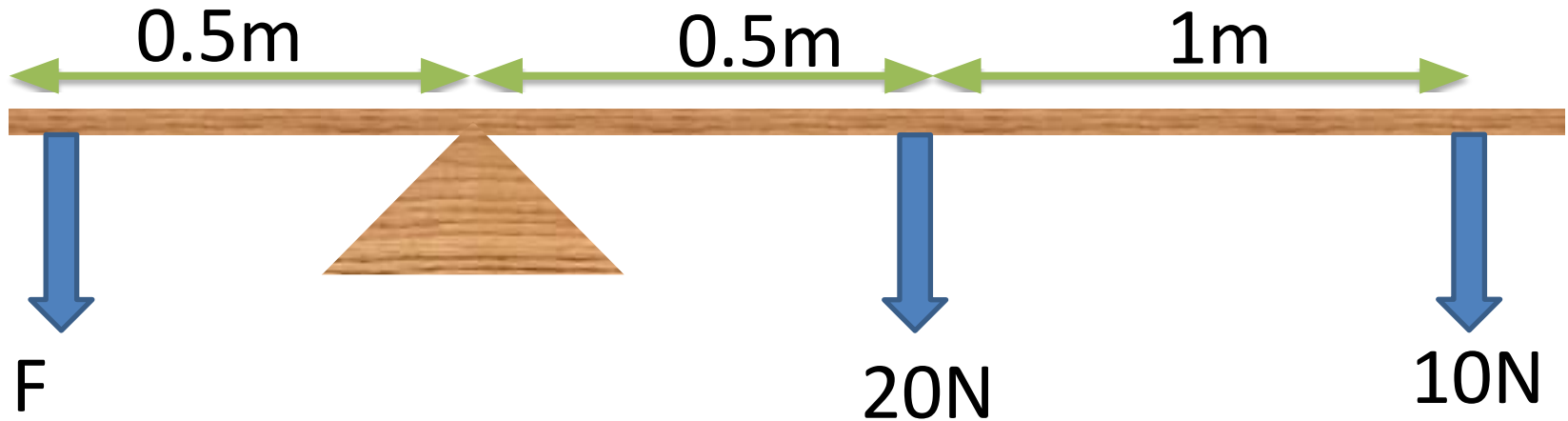
Multiple forces on a beam



- 0.5×28
- $=14$

- $(0.5 \times 10) + ((0.5+1) \times 6)$
- $=5 + 9$
- $=14$

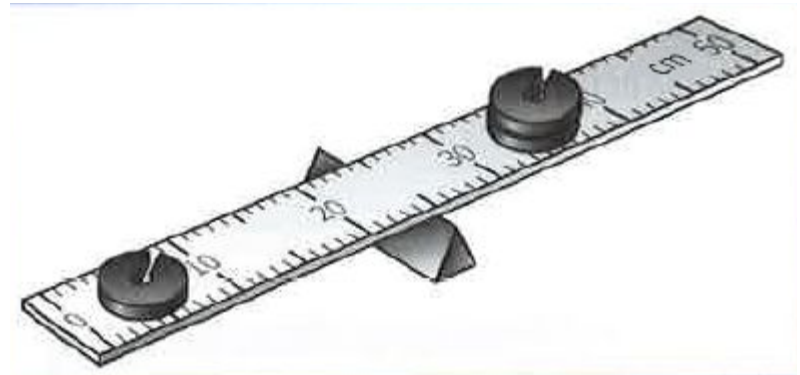
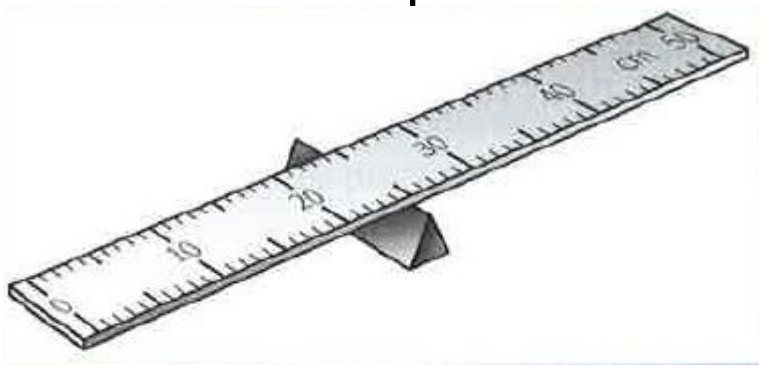
Multiple forces on a beam



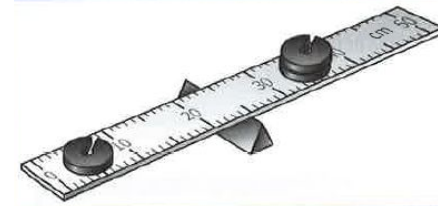
- $0.5 \times F = (0.5 \times 20) + (1.5 \times 10)$
- $0.5 \times F = 10 + 15$
- $0.5 \times F = 25$
- $F = 25 / 0.5$
- $F = 50\text{N}$

Investigating Moments!

- Get a meter ruler and a pivot (triangular block)
 - Balance the ruler at its midpoint on the pivot
1. Check the ruler will balance when you place 50g on each side at equal distances from the pivot. Record the distance.
 2. Put a 50 g mass (weight!) at one end
 - Then work out where you would put a 100g mass to make it balance. Take measurements of the distance of both weights from the pivot.



1. Put a 50 g mass (weight!) at one end
2. Then work out where you would put a 100g mass to make it balance. Take measurements of the distance of both weights from the pivot.
3. Repeat, with 50g mass, and then see where a 200g mass would go the other end.
4. Do a 3rd time with whatever combination of masses
5. Repeat as many times as you have time for.



100g = 1 Newton (on Earth)

Anti-clockwise			Clockwise		
F_1 (N)	D_1 (m)	Moment (Nm)	F_2 (N)	D_2 (m)	Moment (Nm)
1.0			0.5		
1.0			1.0		
1.0			2.0		

Investigating Moments!

- You should know how to describe what you do in this experiment (same as for any expt actually 😊!).
- Write 2 sentences about what you did.
- Then try to look for a pattern in your table of results.
- What do you think a moment is?

Our results

Anti-clockwise Moment			Clockwise Moment		
F_1 (N)	D_1 (m)	$F_1 \times D_1$ (.....)	F_2 (N)	D_2 (m)	$F_2 \times D_2$ (.....)
0.5	0.295		0.5		
1.0	0.24		0.5	0.5	
2.0	0.22		1.0	0.45	
1.0	0.26		0.5	0.47	
5.0	0.1		2.0	<u>?? – answer?</u>	

Learner's Book: page 95



Mark your work using a green pen!

Questions

- 1** **a** the turning effect of a force
- b** $\text{moment} = \text{force} \times \text{distance}$
- c** N m, accept N cm
- 2** **a** A
- b** This force is furthest from the pivot; this force will make the object turn in the correct direction.
- 3** 4×0.12
 $= 0.48 \text{ N m}$
- 4** $\text{moment} = \text{force} \times \text{distance}$
 $\text{force} = \frac{\text{moment}}{\text{distance}}$
 $= \frac{1.8}{0.09}$
 $= 20 \text{ N}$

Mark your work using a green pen!

5 moment caused by Zara = force $\times$ distance
 $= 450 \times 1.5$
 $= 675 \text{ N m}$

To balance, the moments on both sides must be equal.

moment caused by Sofia = force $\times$ distance

$$\text{distance} = \frac{\text{moment caused by Sofia}}{\text{force}}$$

$$= \frac{675}{500}$$

$$= 1.35 \text{ m}$$

